

# Explainable Multimodal Fusion of Histopathological Images and Gene Expression for Breast Cancer Subtype Classification

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## Abstract

Breast cancer is among the most prevalent malignancies worldwide, and accurate identification of molecular subtypes is central to selecting effective treatments for individual patients. Traditional computational approaches rely on either histopathological images or gene expression profiles in isolation, thereby missing the complementary information each modality provides about the tumor microenvironment. We present a multimodal deep learning framework that jointly learns from tissue morphology and transcriptomic data through a bidirectional cross-modal attention mechanism. Our system extracts patch-level visual features using a pre-trained ResNet50, encodes gene expression through a dense projection network, and fuses both modalities via Vision-to-Gene and Gene-to-Vision attention streams embedded within transformer encoder blocks. The framework is evaluated on a matched cohort of 93 patients from the TCGA Breast Invasive Carcinoma dataset using rigorous five-fold stratified cross-validation, classifying tumors as Luminal or Non-Luminal subtypes. We achieve a mean F1-score of 0.914, Matthews Correlation Coefficient of 0.876, and Precision-Recall AUC of 0.976 across folds. To support clinical interpretability, we apply Integrated Gradients, Permutation Importance, and attention output visualization to identify the gene features and tissue patches most informative for each prediction. The source code is available at <https://colab.research.google.com/drive/1a803rPzxw31s0JjBe4XxyhNm02RWTGLW>.

## 1 Introduction

Breast cancer exhibits pronounced molecular heterogeneity, with subtypes including Luminal A, Luminal B, HER2-enriched, and Basal-like tumors differing substantially in their biological behavior, prognosis, and sensitivity to therapeutic agents (Cancer Genome Atlas Network, 2012). Accurately

identifying these subtypes at diagnosis informs treatment selection, making subtype classification a clinically critical step in precision oncology.

In clinical practice, pathologists and oncologists consult both tissue morphology from histology slides and molecular data from genomic assays, yet most computational approaches analyze each modality separately. Histopathological whole-slide images (WSIs) capture rich spatial and morphological detail about the tumor microenvironment but convey little about the molecular drivers of disease (Marusyk et al., 2012). Conversely, RNA-sequencing profiles quantify transcriptome-wide gene activity but carry no spatial information about how cells are organized within tissue (Jackson et al., 2020). Deep learning methods applied to each modality individually therefore face an information ceiling imposed by the limits of a single data view.

Combining these complementary modalities is a natural path toward more accurate and robust classification. Prior multimodal approaches to cancer prediction (Mobadersany et al., 2018; Cheerla and Gevaert, 2019; Chen et al., 2022) have demonstrated that integrating histology and genomic features consistently improves over unimodal baselines. Most existing work, however, relies on simple late fusion through vector concatenation or ensemble averaging (Mobadersany et al., 2018; Cheerla and Gevaert, 2019), which does not explicitly model interactions between features across modalities. Cross-modal attention, introduced in the context of multi-domain sequence modeling (Vaswani et al., 2017), provides a principled mechanism for each modality to selectively attend to relevant features in the other.

We introduce a transformer-based multimodal framework with bidirectional cross-modal attention for breast cancer subtype classification. The framework extracts patch-level visual features from WSIs using a pre-trained ResNet50 (He et al., 2016), pro-

cesses gene expression through a dense encoder, and fuses the two modalities via two complementary attention streams that allow image tokens to query the gene representation and gene tokens to query image patches. To support clinical interpretation, we apply three post-hoc explanation methods that identify which genes and tissue patches drive model decisions.

Our contributions are:

- A multimodal framework combining WSI patch features and full transcriptomic profiles through bidirectional cross-modal attention, enabling explicit inter-modality interaction modeling rather than simple feature concatenation.
- Strong classification performance on the TCGA-BRCA cohort with 93 matched patients, achieving a mean F1-score of 0.914, MCC of 0.876, and PR-AUC of 0.976 across five folds.
- An interpretability analysis using Integrated Gradients, Permutation Importance, and attention output visualization to pinpoint predictive genes and tissue patches.

## 2 Related Work

### 2.1 Deep Learning for Cancer Histopathology

Convolutional neural networks have achieved strong performance on a range of computational pathology tasks, including tumor classification, grade prediction, and mutation detection directly from tissue images (Coudray et al., 2018; Kather et al., 2019; Courtiol et al., 2019). A common paradigm is to tile WSIs into smaller patches, extract patch-level features using networks pre-trained on natural image datasets, and aggregate these features at the patient level. The morphological features captured by such approaches encode relevant information about tumor cellularity, nuclear atypia, and tissue architecture, but they do not reflect the underlying molecular alterations that drive subtype differences (Marusyk et al., 2012).

### 2.2 Multimodal Fusion of Histology and Genomics

Integrating histopathological images with genomic or transcriptomic data has been explored primarily in the context of survival outcome prediction and grade classification. Early approaches relied on late

fusion strategies in which features extracted from independently trained unimodal networks are combined through concatenation before a final classifier. Mobadersany et al. (Mobadersany et al., 2018) concatenated histology image patches with genomic covariates for cancer outcome prediction in glioma. Cheerla and Gevaert (Cheerla and Gevaert, 2019) developed an unsupervised multimodal encoder for pan-cancer prognosis that integrates histology and genomics via concatenation while remaining robust to missing modality data. Shao et al. (Shao et al., 2019) proposed integrative analysis combining pathological images with multi-dimensional genomic data for early-stage cancer prognosis using ordinal feature selection. Huang et al. (Huang et al., 2019) incorporated multi-omics neural networks for survival analysis of breast cancer, demonstrating the value of transcriptome-level information. Chen et al. (Chen et al., 2022) introduced Pathomic Fusion, which models pairwise feature interactions across histology image, cell graph, and genomic modalities through the Kronecker product of gated feature representations, achieving state-of-the-art survival prediction on TCGA data.

A common limitation across these approaches is that most employ late fusion without an explicit mechanism for one modality to selectively attend to relevant features in the other. Our framework addresses this by introducing bidirectional cross-modal attention between the image and gene branches.

### 2.3 Attention Mechanisms and Transformers

The transformer architecture (Vaswani et al., 2017), built on multi-head self-attention, has become a standard tool for modeling long-range dependencies in sequential data. In the multimodal setting, cross-modal attention allows queries from one modality to retrieve information from another, enabling dynamic alignment of features across data types (Baltrusaitis et al., 2019; Zadeh et al., 2017). Gating mechanisms have been used to control the relative contribution of each modality during fusion (Arevalo et al., 2017), reducing the influence of noisy or redundant features before integration.

### 2.4 Interpretability in Medical AI

Interpretability is a practical requirement in clinical settings. Selvaraju et al. (Selvaraju et al., 2017) introduced GradCAM to generate visual saliency maps highlighting image regions relevant to a model’s prediction. Sundararajan et al. (Sun-

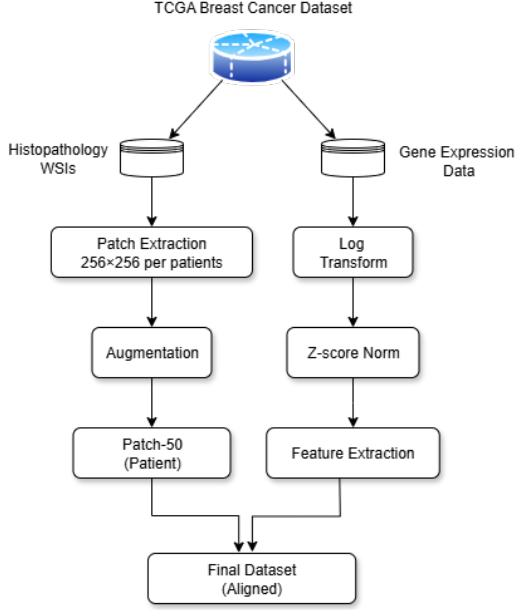


Figure 1: Data preprocessing pipeline for the TCGA-BRCA multimodal cohort. The left branch tiles WSIs into  $256 \times 256$  pixel patches with color jitter augmentation. The right branch applies log1p transformation and z-score normalization to RNA-seq profiles. Both modalities are aligned at the patient level before model training.

dararajan et al., 2017) proposed Integrated Gradients, which attributes model outputs to individual input features through path-integrated gradients and satisfies axiomatic properties of sensitivity and implementation invariance. Permutation Importance provides a model-agnostic alternative that measures the degradation in predictive performance when each feature is individually disrupted, without requiring access to gradients.

### 3 Methodology

#### 3.1 Dataset

We use the Breast Invasive Carcinoma (BRCA) cohort from The Cancer Genome Atlas (TCGA), which provides matched whole-slide images and bulk RNA-sequencing profiles with verified molecular subtype labels (Cancer Genome Atlas Network, 2012). The dataset is available through the GDC Data Portal (<https://portal.gdc.cancer.gov/>).

Subtype labels are grouped into two classes for binary classification: Luminal A and Luminal B are assigned to Class 0, and Basal-like and HER2-enriched are assigned to Class 1. Normal-like samples are excluded to focus on cancer subtype dis-

crimination. After filtering for patients with complete matched data across both modalities, the final cohort comprises 93 patients: 63 Luminal (Class 0) and 30 Non-Luminal (Class 1).

#### 3.2 Data Preprocessing

The full preprocessing pipeline is shown in Figure 1.

**Gene Expression:** Let  $\mathbf{x} \in \mathbb{R}^G$  denote the raw RNA-seq expression vector for a patient, where  $G = 20,318$  after removing genes with zero expression across all samples. Remaining values are log1p-transformed,

$$\tilde{x}_j = \log(1 + x_j), \quad (1)$$

and then standardized via z-score normalization using the training fold statistics. Any residual missing values are imputed with the column mean.

**Histopathological Images:** WSIs are tiled into non-overlapping  $256 \times 256$  pixel patches at  $20\times$  magnification. Up to  $N_p = 50$  patches are sampled per patient; if fewer are available, patches are repeated until the fixed count is reached. Color jitter augmentation is applied during training with brightness and contrast perturbation factors of 0.2, saturation 0.2, and hue perturbation of 0.1.

#### 3.3 Feature Extraction

**Vision:** A ResNet50 (He et al., 2016) pre-trained on ImageNet is used as a fixed feature extractor. The final classification head is removed, and global average pooling applied to the last convolutional feature map yields a  $D_v = 2048$ -dimensional descriptor per patch. For each patient, the  $N_p$  patch descriptors form the feature matrix  $\mathbf{Z} \in \mathbb{R}^{N_p \times D_v}$ .

**Genomics:** The standardized gene expression vector  $\hat{\mathbf{x}} \in \mathbb{R}^G$  is projected to a lower-dimensional embedding through a dense layer followed by batch normalization and dropout:

$$\mathbf{g} = \text{Dropout}(\text{BN}(\text{GELU}(\mathbf{W}_g \hat{\mathbf{x}} + \mathbf{b}_g))), \quad (2)$$

where  $\mathbf{g} \in \mathbb{R}^{D_g}$  with  $D_g = 512$ . The gene feature vector is reshaped to  $\mathbf{G} \in \mathbb{R}^{1 \times D_g}$  to serve as a single token in the subsequent attention module.

#### 3.4 Model Architecture

The overall architecture is shown in Figure 2. It consists of four sequential components: a vision self-attention encoder, a gene projection branch, a bidirectional cross-modal attention fusion module, and a classification head.

### 3.4.1 Vision Self-Attention Encoder

The patch feature matrix  $\mathbf{Z}$  is processed through  $L = 2$  transformer encoder blocks (Vaswani et al., 2017). Each block applies multi-head self-attention (MHSA) to model dependencies across patches, followed by a feed-forward network (FFN) with GELU activation:

$$\mathbf{Z}^{(l)} = \text{LN}\left(\mathbf{Z}^{(l-1)} + \text{MHSA}\left(\mathbf{Z}^{(l-1)}\right)\right), \quad (3)$$

$$\mathbf{Z}^{(l)} = \text{LN}\left(\mathbf{Z}^{(l)} + \text{FFN}\left(\mathbf{Z}^{(l)}\right)\right), \quad (4)$$

where LN denotes layer normalization and MHSA uses  $N_h = 8$  heads with intermediate FFN dimension 512. The output  $\mathbf{Z}^{(L)} \in \mathbb{R}^{N_p \times D_v}$  encodes context-aware patch representations.

### 3.4.2 Bidirectional Cross-Modal Attention

Before attention, both modalities are projected to a shared dimension  $d = N_h \times d_k = 512$ :

$$\mathbf{H}_v = \mathbf{Z}^{(L)} \mathbf{W}_v \in \mathbb{R}^{N_p \times d}, \quad \mathbf{H}_g = \mathbf{G} \mathbf{W}_g \in \mathbb{R}^{1 \times d}. \quad (5)$$

Two complementary attention streams are then computed. In the Vision-to-Gene (V2G) stream, image patch tokens query the gene representation, allowing each patch to retrieve genomically relevant context:

$$\mathbf{H}'_v = \text{LN}(\mathbf{H}_v + \text{MHA}(Q=\mathbf{H}_v, K=\mathbf{H}_g, V=\mathbf{H}_g)). \quad (6)$$

In the Gene-to-Vision (G2V) stream, the gene token queries all patch representations, enabling the molecular feature to attend to the most relevant tissue regions:

$$\mathbf{H}'_g = \text{LN}(\mathbf{H}_g + \text{MHA}(Q=\mathbf{H}_g, K=\mathbf{H}_v, V=\mathbf{H}_v)). \quad (7)$$

Multi-head attention with  $N_h$  heads and per-head key dimension  $d_k$  is defined as:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_{N_h}) \mathbf{W}^O, \quad (8)$$

$$\text{head}_i = \text{Attention}(Q \mathbf{W}_i^Q, K \mathbf{W}_i^K, V \mathbf{W}_i^V), \quad (9)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \quad (10)$$

### 3.4.3 Pooling and Fusion

Global average pooling reduces the attended image sequence to a fixed-size vector:

$$\mathbf{u}_v = \frac{1}{N_p} \sum_{i=1}^{N_p} \mathbf{H}'_{v,i} \in \mathbb{R}^d. \quad (11)$$

The attended gene token is squeezed to  $\mathbf{u}_g = \text{squeeze}(\mathbf{H}'_g) \in \mathbb{R}^d$ .

The two representations are concatenated to form the joint embedding:

$$\mathbf{c} = [\mathbf{u}_v; \mathbf{u}_g] \in \mathbb{R}^{2d}. \quad (12)$$

A two-layer MLP with GELU activations, batch normalization, and dropout produces the binary classification logit:

$$\hat{y} = \sigma(f_\theta(\mathbf{c})) \in [0, 1], \quad (13)$$

where  $\sigma$  is the sigmoid function and  $f_\theta$  is the classification MLP.

### 3.5 Explainability

**Integrated Gradients** (Sundararajan et al., 2017):

For a gene feature  $x_i$  with a zero-vector baseline  $\mathbf{x}' = \mathbf{0}$ , the attribution score is approximated using  $m = 50$  interpolation steps:

$$\text{IG}_i(\mathbf{x}) \approx (x_i - x'_i) \cdot \frac{1}{m} \sum_{k=1}^m \frac{\partial F(\mathbf{x}' + \frac{k}{m}(\mathbf{x} - \mathbf{x}'))}{\partial x_i}, \quad (14)$$

where  $F$  is the predicted probability. Genes with large absolute IG values are those whose expression levels most influence the model output for a given sample.

**Permutation Importance:** For each gene  $j$ , the importance score is the accuracy drop when the values of that gene are randomly shuffled across samples:

$$\text{Imp}_j = \text{Acc}(\mathbf{X}) - \text{Acc}(\tilde{\mathbf{X}}_j), \quad (15)$$

where  $\tilde{\mathbf{X}}_j$  denotes the dataset with column  $j$  permuted. A positive score indicates that disrupting gene  $j$  reduces accuracy, confirming its predictive role.

**Attention Output Visualization:** The hidden-state outputs of the V2G and G2V cross-attention layers are extracted at inference time to produce heatmaps showing how each attention head distributes its response across image patches and gene tokens.

### 3.6 Training Procedure

The model is trained with binary cross-entropy loss and the Adam optimizer with learning rate  $\eta_0 = 10^{-5}$  and batch size 8. An exponential learning rate schedule is applied after epoch 10:

$$\eta_t = \eta_0 \cdot e^{-0.1(t-10)}, \quad t > 10. \quad (16)$$



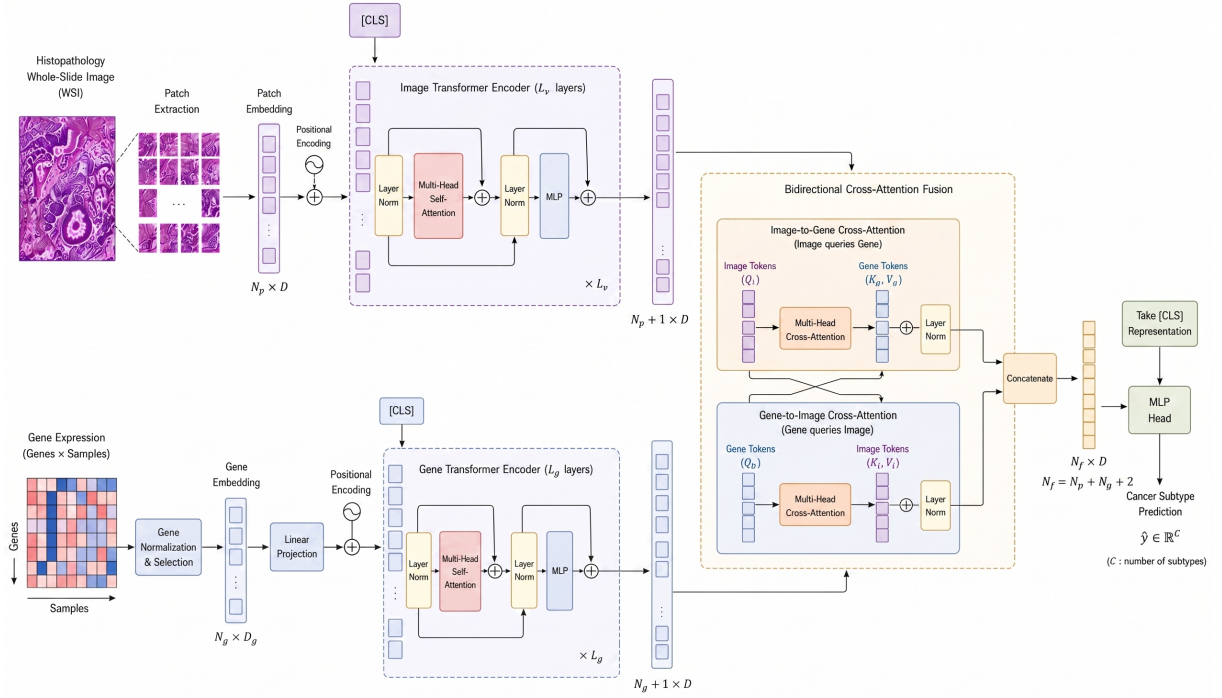


Figure 2: Architecture of the proposed multimodal framework. The upper branch encodes WSI patch sequences through a vision transformer encoder. The lower branch projects gene expression to a compact token representation. Bidirectional cross-modal attention (center) enables image tokens to query the gene token (V2G) and the gene token to query all image tokens (G2V). The attended representations are pooled, concatenated, and classified by an MLP head.

Training runs for a maximum of 35 epochs, with early stopping based on validation loss with patience 10. The total model has approximately 52.3 million trainable parameters. Five-fold stratified cross-validation at the patient level ensures that no patient appears in both training and validation within any fold.

## 4 Experimental Setup

### 4.1 Dataset Details

The TCGA-BRCA cohort originally contains 1,247 patients with matched WSIs and RNA-seq profiles. After removing samples without complete data alignment across both modalities and excluding Normal-like cases, 93 patients remain: 63 Luminal (Class 0) and 30 Non-Luminal (Class 1). The class ratio reflects the natural distribution of molecular subtypes in the TCGA cohort. For each patient, up to 50 image patches of size  $256 \times 256$  pixels are extracted and at most  $G = 20,318$  gene expression features are retained after zero-expression filtering.

### 4.2 Evaluation Metrics

We report three complementary metrics for each fold and their mean and standard deviation across five folds.

**F1-Score** is the harmonic mean of precision and recall for the positive (Non-Luminal) class, emphasizing the balance between false positives and false negatives under class imbalance.

**Matthews Correlation Coefficient (MCC)** provides a balanced single-number summary of binary classification quality that accounts for all four cells of the confusion matrix:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \quad (17)$$

**Precision-Recall AUC (PR-AUC)** summarizes classification quality over all decision thresholds, which is particularly informative under class imbalance.

| Fold | F1                  | MCC                 | PR-AUC              |
|------|---------------------|---------------------|---------------------|
| 1    | 1.0000              | 1.0000              | 1.0000              |
| 2    | 0.9231              | 0.8895              | 0.9062              |
| 3    | 0.8000              | 0.7161              | 0.9742              |
| 4    | 0.9231              | 0.8864              | 1.0000              |
| 5    | 0.9231              | 0.8864              | 1.0000              |
| Mean | $0.9138 \pm 0.0642$ | $0.8757 \pm 0.0909$ | $0.9761 \pm 0.0364$ |

Table 1: Five-fold cross-validation results on the TCGA-BRCA cohort. Mean and standard deviation are computed across folds.

### 4.3 Implementation Details

All experiments were conducted on Google Colab with an NVIDIA V100 GPU. The framework is implemented in TensorFlow/Keras (Python 3.11). ResNet50 weights are initialized from ImageNet pre-training and held fixed throughout. The random seed is fixed to 39 for reproducibility. Five-fold stratified cross-validation is implemented using StratifiedKFold with shuffle enabled and seed 42.

## 5 Results and Analysis

### 5.1 Classification Performance

Table 1 reports the per-fold classification metrics and their average over five folds.

The framework achieves a mean F1-score of 0.914 and MCC of 0.876, indicating that the model consistently separates Luminal from Non-Luminal subtypes with few misclassifications. The PR-AUC of 0.976 confirms reliable discrimination at all decision thresholds, which is particularly meaningful given the 2:1 class imbalance in this cohort. Fold 1 achieves perfect classification of all 19 validation patients, with confusion matrix  $\begin{bmatrix} 13 & 0 \\ 0 & 6 \end{bmatrix}$ . The only source of error across all five folds is false-positive predictions (Luminal patients classified as Non-Luminal); every fold records zero false negatives, meaning no Non-Luminal patient is misclassified as Luminal.

Fold 3 shows the greatest performance drop (F1 = 0.800, MCC = 0.716), driven by three false-positive predictions in a validation set of 19 patients. This fold converged at epoch 20 via early stopping, suggesting the validation partition in that fold was more challenging than average. The relatively low standard deviation across folds (F1  $\sigma = 0.064$ ) demonstrates that the learned representations are stable across different patient partitions.

### 5.2 Explainability Analysis

Integrated Gradients identify gene-level attribution patterns per sample, highlighting expression features that push the model toward or away from the Non-Luminal class. At the population level, the top-attributed genes from IG analysis are consistent with known oncological markers that distinguish basal-like and HER2-enriched tumors from Luminal subtypes, including genes implicated in cell proliferation, receptor tyrosine kinase signaling, and epithelial-to-mesenchymal transition.

The Gene-to-Vision (G2V) attention output maps show that the gene token concentrates its attention on patch regions displaying high nuclear density, cellular atypia, and irregular tissue architecture, features recognized by pathologists as hallmarks of aggressive non-luminal breast cancer subtypes. This cross-modal alignment suggests that the bidirectional attention mechanism has learned meaningful associations between molecular profiles and morphological phenotypes, rather than exploiting either modality independently.

The permutation importance analysis was computed on a randomly sampled subset of 200 genes out of 20,318. No genes in this subset showed non-zero importance scores, meaning that disrupting any individual gene in the sample did not produce a measurable drop in accuracy. This outcome reflects the limitations of evaluating less than 1% of the feature space: genes with predictive value outside the sampled 200 are not assessed, and disrupting a single gene in a dense high-dimensional representation may not visibly degrade accuracy on its own. A full sweep over all genes, or grouped permutations by biological pathway, is needed to draw conclusions from this method.

### 5.3 Training Dynamics

The learning rate schedule and early stopping produce stable convergence across all folds. Folds 1, 2, and 4 complete all 35 training epochs, while folds 3 and 5 trigger early stopping at epoch 30, with their best checkpoints restored from epoch 20. Fold 1 reaches perfect validation accuracy (1.000) for the first time at epoch 14 and maintains it for the remaining epochs. The relatively slow initial learning rate of  $10^{-5}$  prevents large gradient steps from disrupting the pre-trained ResNet50 features while allowing the transformer and cross-modal attention components to learn from scratch.

## 6 Conclusion

We presented a multimodal deep learning framework that integrates histopathological patch features and bulk gene expression data for breast cancer subtype classification through bidirectional cross-modal attention. By allowing each modality to attend to the other, the model learns coordinated representations that capture correspondences between tissue morphology and transcriptomic patterns. Evaluated on the TCGA-BRCA cohort with five-fold cross-validation, the framework achieves a mean F1-score of 0.914, MCC of 0.876, and PR-AUC of 0.976, with high consistency across folds. The integrated explainability analysis provides per-sample and population-level insight into which genes and tissue regions drive classification, supporting the interpretability requirements of clinical decision support systems. Future work will explore larger matched cohorts, multi-class subtype discrimination, and ablation studies comparing unimodal baselines with the full multimodal system.

## Limitations

The main limitation of this work is the small cohort size. After filtering for complete matched data, only 93 patients are available for training and evaluation. Although five-fold cross-validation provides a reliable performance estimate, the model is trained on approximately 74 patients per fold, which may limit the complexity of representations that can be reliably learned. A larger matched dataset would allow more thorough investigation of subtype-specific patterns and more robust generalization estimates.

Second, the current evaluation does not include comparisons against unimodal baselines or alternative fusion strategies such as simple concatenation. Such comparisons would quantify the specific contribution of bidirectional cross-modal attention over simpler integration methods and are left for future work.

Third, the permutation importance analysis was run on a random subset of 200 genes (roughly 1% of the full feature set) due to computational constraints, and returned no genes with non-zero importance in that subset. This outcome means the analysis cannot be used to rank genes by predictive importance; a full sweep or pathway-grouped permutations is needed before any conclusions can be drawn from this explainability method.

Finally, the GradCAM-based patch importance analysis requires specifying a target layer with a three-dimensional output aligned to the patch dimension. Automatic layer selection was not always successful in the current implementation, limiting the patch-level spatial explanations to cases where a suitable layer could be identified.

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